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Full Length Article

Trust my IDS: An explainable AI integrated deep learning-based transparent threat detection system for industrial networks

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ABSTRACT

Industrial networks are vulnerable to various cyber threats that can compromise their Confidentiality, Integrity, and Availability (CIA). To counter the increasing frequency of such threats, we designed and developed an Explainable Artificial Intelligence (XAI) integrated Deep Learning (DL)-based threat detection system (XDLTDS). We first employ a Long-Short Term Memory-AutoEncoder (LSTM-AE) to encode IIoT data and mitigate inference attacks. Then, we introduce an Attention-based Gated Recurrent Unit (AGRU) with softmax for multiclass threat classification in IIoT networks. To address the black-box nature of DL-based IDS, we use the Shapley Additive Explanations (SHAP) mechanism to provide transparency and trust for the system's decisions. This interpretation helps SOC analysts understand why specific events are flagged as malicious by the XDLTDS framework. Our approach reduces the risk of sensitive data and reputation loss. We also present a Software-Defined Networking (SDN)-based deployment architecture for the XDLTDS framework. Extensive experiments with the N-BaIoT, Edge-IIoTset, and CIC-IDS2017 datasets confirm the effectiveness of XDLTDS against existing frameworks in addressing modern cybersecurity challenges and protecting industrial networks.

1. Introduction

The Internet of Things (IoT) is a transformative communication approach representing a mutually connected pattern of physical objects, devices, sensors, and protocols to formulate a smart data-exchanging framework (Rathee et al., 2022). IoT systematically enables an automated networking architecture where the sensors interlink the participant nodes, and the communication protocol regulates the data transmission, consequently mitigating the need for direct human interaction (Ashraf et al., 2022). The concept is widely acknowledged, resulting in extensive growth of IoT applications in every sector, including smart homes, smart cities, smart healthcare, smart agriculture, smart transportation, smart monitoring systems, etc. The industrial domain is more likely privileged by IoT and reveals a vast variety of IoT applications in the industrial scenario, referred to as the Industrial Internet of Things (IIoT). Over the past decade, IIoT has significantly revolutionized the typical paradigm for industrial systems by introducing a fusion of innovative technologies into conventional industrial processes (Adil et al., 2022). The phenomenon is interpreted by the literature term Industry 4.0, yielding numerous benefits, e.g., smart manufacturing, communication reliability, resource optimization, predictive maintenance, increased efficiency, enhanced safety measures, and many more (Chiu et al., 2022). The increasing demands of IIoT and the flourishing circle of its applications experience the involvement of heterogeneous communication nodes interlinked by cyber links over the internet. Such multidimensional heterogeneous connectivity becomes a potential ground for bad actors to perform adversarial activities to compromise the integrity of the system. The examples must include zero-day exploits, Cross-Site Scripting (XSS), Structured Query Language (SQL), and Man-in-the-Middle (MITM) attack categories (Li et al., 2022). Counting these categories, the IIoT systems may also become vulnerable to a broader range of other security challenges, such as Denial of Service (DoS) attacks, crypto-jacking attacks, replay attacks, side-channel attacks, device spoofing attacks, cloud-based attacks (Chen et al., 2020), etc. Along with the security breaches, the threats also prohibit IIoT systems from meeting operational sustainability, efficient scalability, and regulatory compliance. Such circumstances flag an alarming situation and require significant security countermeasures to be adopted.

Artificial Intelligence (AI) provides a productive insight into developing security solutions to address the mentioned challenges. Deep

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Learning (DL) is an AI-driven pathway increasingly implied to design more sophisticated security frameworks for IIoT. In recent years, DL-empowered Intrusion Detection Systems (IDS) have attained considerable attention since their complex computational cycles in decision-making, leading to substantial filtering between normal and suspicious scenarios (Lo et al., 2022). DL-based IDS is initially trained on a dataset containing the impressions of threats and then employed to identify malicious activities from regular ones. Undoubtedly, DL-driven IDS are a remarkable choice for the timely detection of malicious entities in IIoT communications. However, there is still ample room for improvement to boost the reliability and trustworthiness of these security methods (Javeed et al., 2023b). Traditional DL approaches adopt a black-box strategy where all the internal processing of algorithms is hidden, making it challenging for humans to understand their decision-making process. While working in sensitive industrial environments, it becomes crucial for security analysts to consider the logic behind declaring a specific activity as malicious (Ahmed et al., 2022). This context raises unaddressed questions regarding the steps involved in the decision-making processes of DL approaches, prompting the need for some transparent solutions. Explainable Artificial Intelligence (XAI) is an emerging approach that refers to an absolute package of methods and techniques to interpret the decision-making process of DL algorithms (Rawal et al., 2021). XAI explores the black-box concept and explains the steps contributing to the decision-making process, providing humans with a clear understanding of the logic employed to flag a particular event as malicious. XAI facilitates conventional DL algorithms with transparency, interpretability, trustworthiness, and reliability. XAI also makes the DL approaches accountable for making certain decisions, reducing the probabilities of inefficiencies and biased decision-making (Kök et al., 2023). While addressing high-stakes industrial environments, it becomes comparatively more indispensable to interpret the decisions taken by automated systems concerning regulatory compliance. Integrating XAI in DL-driven IDS is an impressive idea for comprehending decision-making and transparently considering the intrusion detection process.

1.1. Problem statement and research questions

IIoT is revealing manifestation in every sphere of our lives and its application circle is remarkably expanding to assist disparate aspects of our lifestyle. While providing several benefits, it is also vulnerable to a myriad of cyber threats that can lead to severe consequences, i.e., equipment damage, production disruptions, data breaches, and data loss. A typical IoT-based industrial network may be a susceptible victim to a variety of cyber-attacks, i.e., botnet attacks, spoofing attacks, Denial of Service (DoS), eavesdropping, and Distributed DoS (DDoS) attacks (Zainudin et al., 2022). Consequently, the IIoT demands an efficient Threat Detection System (TDS) to maintain its functionality and integrity. Designing a TDS for the IIoT networks is of paramount importance to ensure the privacy, stability, and security of its critical processes. Numerous researchers have suggested ML and DL-based IDSs in recent years to protect the IIoT network from such virulent threats and attacks (Javeed et al., 2023a). However, such models are frequently prone to overfitting and are considered black boxes due to their lack of interpretability. Thus, the IIoT is still in dire need of an efficient, transparent, and resilient framework to secure its network against sophisticated threats. Therefore, we will consider the following Research Questions (R) in this research work:

- **(R1): How to develop an efficient and resilient IDS to detect the sophisticated threat in such a network with high accuracy and low false alarm rate?** The existing signature-based IDSs mostly rely on familiar attack signatures and have notable false negative and false positive rates, which limits their capability of detecting unknown threats. Although these signature-based IDSs are known to be effective, they may be outsmarted by the attackers using various ways. Consequently, it makes them impractical for deployment.
- **(R2): How to provide a solution for IIoT's heterogeneous nature and develop a scalable and centralized IDS that does not overburden the resource-constrained IIoT devices?** The IIoT networks have distinctive properties and require different threat detection mechanisms than the traditional networks. It consists of a considerable variety of resource-constrained and heterogeneous devices following different protocols, where each device requires diverse security measures. Further, IIoT networks frequently scale quickly with new devices constantly being added.
- **(R3): How to develop a transparent Threat detection System and interpret its decision for enhanced Trust?** The lack of transparency of the DL-based IDSs is one of the reasons for their vigilant adoption in critical systems. The security analysts consider such model black boxes as they lack the interpretation of their decisions. Understanding the rationale behind a model decision-making process promotes trust, and confidence, enabling the Security Operation Center (SOC) analysts to confirm that it is efficiently and securely doing its job.

1.2. Contribution

The following are the contributions of this research work.

- **To address R1,** we design a DL-based TDS to protect IIoT networks against evolving threats. The proposed framework is developed using Long-Short Term Memory-Auto Encoder (LSTM-AE) to encode the IIoT network data in order to prevent inference attacks. After that, we employ an Attention-based Gated Recurrent Unit (AGRU) with softmax for multiclass classification of attacks in such a network.
- **To address R2,** we proposed an SDN-based deployment architecture for the proposed TDS. We placed the proposed TDS into the CP of the SDN for the following reasons. Firstly, it empowers an adaptive, dynamic, and scalable security mechanism, which means we do not need to major architectural changes with the expansion of the network. Secondly, it aids Security Operation Center (SOC) analysts in identifying threats efficiently with low false alarm rates by minimizing unnecessary alerts, which minimizes the potential damage to the network. Thirdly, it optimizes resource usage and does not overburden the resource-constrained devices, thus ensuring the overall reliability and security of the IIoT networks.
- **To address R3,** we employ the SHAP mechanism of the XAI to interpret the feature of the datasets that primarily contributed to the predictions of the proposed IDS. This explainability of the decision will help the SOC analysts to better understand and analyze the decision. In addition, it will promote trust and reliability of the proposed framework. Consequently, our proposed framework will allow industries to defend against sophisticated threats to reduce the risk of sensitive data loss and reputation loss.

The rest of the paper is organized as follows; Section 2 discusses the relevant literature. Section 3 presents the proposed mechanism. Section 4 describes the experimental setup with evaluation metrics. Section 5 discusses the results and performance of the proposed XDLTDS framework. Section 6 discusses the XAI-based Interpretation. In Section 7, we provide a discussion. Finally, Section 8 concludes the paper.

2. Literature overview

Industrial network security has grabbed the attention of researchers for the past few years, and they proposed various threat detection mechanisms as shown in Table 1. For example, researchers in Ding et al. (2023) proposed a DL-based model named DeepAK-IoT for cyber threat detection. Their proposed model comprises three blocks, i.e., the

Table 1
Existing literature overview.

Ref	Year	Model	Strength	Limitation
Ding et al. (2023)	2023	DeepAK-IoT	Rapid in against attack detection.	Lack interpretation of the decision-making process.
Taher et al. (2023)	2023	IHHO-NN	Efficient detection of multi-class botnet attacks.	Their proposed IDS lack a deployment architecture.
Illy et al. (2022)	2022	DNN	Their proposed approach provide an efficient classification accuracy. Further, an SDN-based deployment architecture is also provided.	Vulnerable to Inference attacks.
Khan et al. (2022b)	2022	LSTM autoencoder	Their proposed techniques have a better detection accuracy of 97.95% than the other models using the same dataset.	Unable to identify the attack classes precisely.
Shukla et al. (2023)	2023	RF	Their proposed model is computationally faster than the DL-based methods.	The detection accuracy is 63%, which is not considered promising in the Industrial ecosystem.
Santhadevi and Janet (2023)	2023	HSDL	The authors achieved an efficient threat detection accuracy.	The model is black-box.
Telikani et al. (2022)	2022	CNN-SAE	The proposed framework has the ability to increase the detection ratio of low-frequency intrusions.	The proposed IDS has high transmission overhead with low accuracy.
Lilhore et al. (2023)	2023	CNN–LSTM	Transfer learning is used to improve the performance of IDS using knowledge transformation.	Low accuracy and precision with high loss.
Khacha et al. (2022)	2022	CNN–LSTM	The model achieved good accuracy with low false alarm rate.	Lack deployment architecture and unable to identify attack classes.
Xu et al. (2023)	2023	EE-GNN	The authors classify network traffic into attack and normal by extracting relevant features from data to design graph networks and characterize connectivity properties using an adjacency matrix between device nodes.	Black box model with low detection accuracy.
Tareq et al. (2022)	2022	Inception Time	A sliding window approach is used to improve the detection accuracy.	Lacks explanation of the decision and deployment architecture of IDS.
Friha et al. (2023)	2023	2DF-IDS	Their proposed approach eliminates the risk of a single point of failure.	Low detection accuracy and black box model.

RSR block comprising CNN, the TRB block comprising GRU and LSTM, and the DB block comprising softmax. The authors utilized TON-IoT, Edge-IIoT, and UNSW-NB15 datasets for performance evaluation of their IDS and got a detection accuracy of 94.96% using the Edge-IIoT dataset and 90.57% using the TON-IoT and 98.41% UNSW-NB15 datasets. Researchers in Taher et al. (2023) proposed an attack detection mechanism to address trending cyber attacks in IIoT, such as doS attacks, botnet attacks, and flooding attacks. By inspiring with the base model Deep Neural Network (DNN), the authors proposed an Improved Harris Hawks Optimization Algorithm for Neural Networks (IHHO-NN) to categorize attack classes efficiently. They trained their model using the N-BaIoT dataset and achieved reliable resilience against the mentioned attacks upon the performance evaluation of their model. The authors in Illy et al. (2022) devised a novel Intrusion Detection and Prevention System (IDPS) by integrating DL and SDN. The proposed model aims to interrogate suspicious events in high-risk smart factory networks. The DNN classifier is acquired for the intrusion detection mechanism, whereas SDN is interlinked to provide a supplementary execution architecture to the proposed model. Researchers used two datasets for training purposes, i.e. WUSTL-IIoT-2018 and NSL-KDD.

Authors in Khan et al. (2022b) designed another attack detection framework for industrial control systems. The designed model targets emerging infrastructural-based attack categories, e.g., seismic attacks, flame attacks, duaqu attack classes, etc. The model contains an LSTM autoencoder and is trained on two datasets, i.e., the gas pipeline dataset and the UNSW-NB15 dataset. The system's performance is examined to get an idea about the model's accuracy, and the system achieved 97.95% accuracy for gas pipelines and 97.62% on UNSW-NB15 datasets. The researchers in Shukla et al. (2023) suggested another intelligent, unsupervised intrusion detection scheme for industry 4.0 communications. The proposed attack identification framework is leveraged with a Random Forest (RF) classifier for efficient feature extraction that further helps classify various attack classes unsupervised. The training of the model is performed on the UNSW-NB15 dataset and consists of instances of relevant attacking activities in smart industrial environments. The model depicts reliable performance to detect adversarial activities on time. However, the false positive rates were relatively high. The authors in Santhadevi and Janet (2023) proposed a Hybrid Stacked Deep Learning (HSDL) model-based IDS. The authors utilized Stacked LSTM and Deep Vectorized CNN to design their proposed model, where the Deep Vectorized CNN extracts the features from the data and Stacked LSTM is used for classification purposes.

Researchers in Telikani et al. (2022) contributed to securing IIoT against cloud-based attacks and proposed a DL-based threat detection system. The system is facilitated by CNN and Staked Autoencoder (SAE) classifiers and combined with fog computing to establish a uniform execution architecture. The system is trained on the UNSW-NB15 and ToN-IoT datasets and is evaluated to examine its threat detection capabilities. Researchers in Lilhore et al. (2023) suggested an intelligent threat classification model (CNN–LSTM) using the combined strength of CNN and LSTM classifiers. The model intends to identify common adversaries in industry 4.0-based smart factories. The authors trained their model on two remarkable datasets, ToN-IoT and UNSW-NB15. The system is investigated to evaluate its performance regarding attack detection accuracy, where the obtained results project 94.4% accuracy on the ToN-IoT dataset and 91.02% on the UNSW-NB15 dataset. Authors in Khacha et al. (2022) designed an anomaly detection model by acquiring CNN and LSTM classifiers. The system aims to address DoS, MiTM attacks, and malware attacks in smart industrial networks. The researchers trained the proposed model on an Edge-IIoT set that consists of identical instances of such attack categories. The authors in Xu et al. (2023) acquired a Graph Convolutional Network (GCN) to formulate an Exploiting Edge (EE) features-based attack detection framework (EE-GNN). The objective is to filter abnormal traffic streams from the normal ones in smart industrial communications. The proposed scheme is trained on two significant datasets, and performance is evaluated to explore the attack detection accuracy of the proposed model. The obtained results validate 83.79% and 64.71% accuracy on BoT–IoT and ToN-IoT datasets, respectively. The authors of Tareq et al. (2022) proposed DenseNet and inception time-based approach to detect cyber threats. They employed three publicly available datasets,

i.e., Edge-IIoT, ToN-IoT, and UNSW-NB15 for experimentation purposes and evaluated their proposed threat detection framework by employing standard evaluation metrics. The authors achieved 94.94% accuracy using the Edge-IIoTset dataset, 98.40% using the UNSW-NB15 dataset, and an accuracy of 99.9% using the ToN-IoT dataset. The researchers in Friha et al. (2023) designed a Decentralized, Differentially Private Intrusion Detection System (2DF-IDS) for IIoT communication scenarios. The proposed model employs the Federated Learning (FL) technique and is equipped with DNN. The authors used the Edge-IIoTset dataset for training end valuation purposes. The system's performance is comprehensively examined, with a projected 94.37% attack detection accuracy.

Furthermore, the authors of Zhang et al. (2024) proposed a framework namely "HPPELM" for intrusion detection in the Internet of Vehicles. The authors trained and evaluated their proposed framework by using NSL-KDD and CIC-IDS2017 datasets and achieved a 91.02% accuracy on NSL-KDD and 94.67% accuracy on CIC-IDS2017 datasets. In Hasan et al. (2023), the authors proposed an XAI-integrated DL-based IDS for IIoTs. They employed an ensemble DL model for intrusion detection purposes while using XAI to interpret their model's decision-making process. The authors utilized the publicly available ToN_IoT dataset to evaluate their model and achieved efficient outcomes for binary classification. Similarly, the authors in Keshk et al. (2023) designed an LSTM-based IDS for IoT networks along with the SHAP mechanism of XAI for interpretation purposes. The authors validated the performance of their proposed IDS by using three publicly available datasets, i.e., NSL-KDD, UNSW-NB15, and ToN_IoT. The authors of Khan et al. (2022a) designed an SDN-enabled DNNGRU-based IDS for fog-to-things paradigms. They utilized the control plane of SDN to deploy their proposed IDS. The authors evaluated their IDS using the publicly available CSE-CICIDS2018 dataset and achieved efficient outcomes. In Arthi et al. (2024), the authors developed an intelligent SDN-enabled secure framework for smart healthcare systems. The authors utilized ML and DL-based models, i.e., DNN and SVM to detect threats in the healthcare data and achieved precision, recall, and f1-score of 96%, 96%, and 97%. Finally, the authors of Oseni et al. (2022) developed an IDS for IoT-enabled transportation systems using DL and XAI. The authors utilized the CNN model for threat detection and the SHAP mechanism of XAI to provide global and local explanations of the features that contributed to the IDS's decision. They evaluated their IDS using all the standard evaluation metrics and achieved an accuracy of 99.15% and 90.55% for binary and multi-class classifications.

Most of the existing solutions proposed by the authors either achieved low detection accuracy or a low detection rate. Furthermore, these frameworks are considered black boxes by researchers and security experts because they lack the transparency and interpretability of their decisions. Moreover, the current solutions lack deployment architecture and the heterogeneous and resource-constrained nature of IIoT devices remains a challenging issue. This gap suggests a major research opportunity to develop a unified cybersecurity paradigm that not only leverages DL for security but also incorporates a deployment architecture to optimize resource usage that does not overburden the resource-constrained IIoT devices. Further, the integration of an XAI-based mechanism is also required for transparency and accountability of the decision-making processes of such a framework. The current literature's emphasis on discrete elements highlights the possibility of a comprehensive strategy that might overcome the present IIoT security limitations in terms of trust, transparency, and explainability. Consequently, an efficient, interpretable, robust, and transparent mechanism is still desperately needed for the IIoT to protect its network from sophisticated attacks.

3. Proposed mechanism

In this section, we provide the complete details of our proposed approach. First, we discuss the XAI-integrated DL-based threat detection module. Second, we present details about the Deployment Architecture.

3.1. XAI integrated DL-based framework (XDLTDS) for privacy preservation and security

3.1.1. System architecture

The XDLTDS framework is a combination of several processes and modules. Fig. 1 presents the working architecture of the proposed XDLTDS scheme. First, we use three publicly available datasets, i.e., N-BaIoT, Edge-IIoTset, and CIC-IDS2017 datasets to train and evaluate our proposed model. In particular, the proposed framework is developed by using LSTM-AE to encode IIoT data in order to avoid inference attacks. We further employed AGRU with softmax for multiclass attack classification. In addition, we employed XAI to interpret the top contributed features of these datasets to provide an explanation of the decision of the proposed XDLTDS framework. Our proposed framework aims to aid the SOC analysts to better understand and analyze the decision. In addition, it will promote trust and reliability by helping SOC analysts verify their decisions, defend the industrial network against cyber threats, and maintain the security of the organization by reducing the risk of data loss.

3.1.2. Long-short Term Memory Autoencoder (LSTM-AE)

The LSTM-AE has two parts, i.e., encoder and decoder. The encoder converts the input (D_t) through a deterministic affine mapping function and nonlinearity into hidden representation (Q_t) using Eq. (1) (Kumar et al., 2021):

$$Q_t = \mathcal{F}(\mathcal{W}D_t + \mathcal{B}) \tag{1}$$

where $\mathcal{W}$ is the weight between the input D_t and hidden representation Q_t and bias $\mathcal{B}$ represents the $\mathcal{W}$ bias. The decoder then reconstructs the output $\hat{D}_t$ using Q_t;

$$\hat{D}_t = \mathcal{F}'(\mathcal{W}'D_t + \mathcal{B}') \tag{2}$$

where $\mathcal{W}'$ is the weight between input D_t and output $\hat{D}_t$. $\hat{D}_t$ is the reconstruction of D_t. The primary goal of the training of the AE is to minimize the Reconstruction Error ($\mathcal{RE}$), which is possible by lowering the cost function $\mathcal{M}$:

$$\mathcal{M} = \frac{1}{\mathcal{P}} \sum_{i=1}^{\mathcal{P}} \mathcal{L}[D_t, \hat{D}_t] \tag{3}$$

where the input features number is denoted by $\mathcal{P}$, D_t is the tth feature of input, and output $\hat{D}_t$ relates to the reconstructed feature. The $\mathcal{L}[D_t, \hat{D}_t]$ represents the input D_t reconstruction error. This can be quantified by Cross-Entropy (CE) or Mean Square Error (MSE). The MSE of D_t, (t = 1, 2, 3, ……, $\mathcal{P}$) and $\hat{D}_t$, (t = 1, 2, 3, …., $\mathcal{P}$) is used in this analysis. Subsequently, $\mathcal{L}[D_t, \hat{D}_t]$ can be calculated as (Kumar et al., 2021).

$$\mathcal{L}[D_t, \hat{D}_t] = \|D_t - \hat{D}_t\|^2 \tag{4}$$

The model exploits the LSTM in conjunction with the AE, meaning that LSTM handles the encoding and decoding. The most significant components of the training data are forced to be recognized by the LSTM-AE by restricting the latent space to have a smaller dimension than the input. In short, to prevent inference attacks, when the attacker seeks to ascertain if a given input is a component of the model's training data before attempting to learn information about end users, we utilized LSTM-AE to transform the IIoT network data into a new format.

3.1.3. Attention-based Gated Recurrent Unit (AGRU)

GRU is an enhanced version of RNN. Specifically, it is designed to overcome the limitations of the RNN. GRU incorporates a gating technique that aids in controlling the information flow inside the network. The GRU can capture long-range dependencies in sequences more successfully than conventional RNNs because of its gating mechanism. It introduces two gates, i.e., an Update Gate ($\mathcal{U}_t$) and Reset Gate ($\mathcal{R}_t$) to address the vanishing gradient issue that plagues a normal RNN (Dey and Salem, 2017). These two gates are responsible for

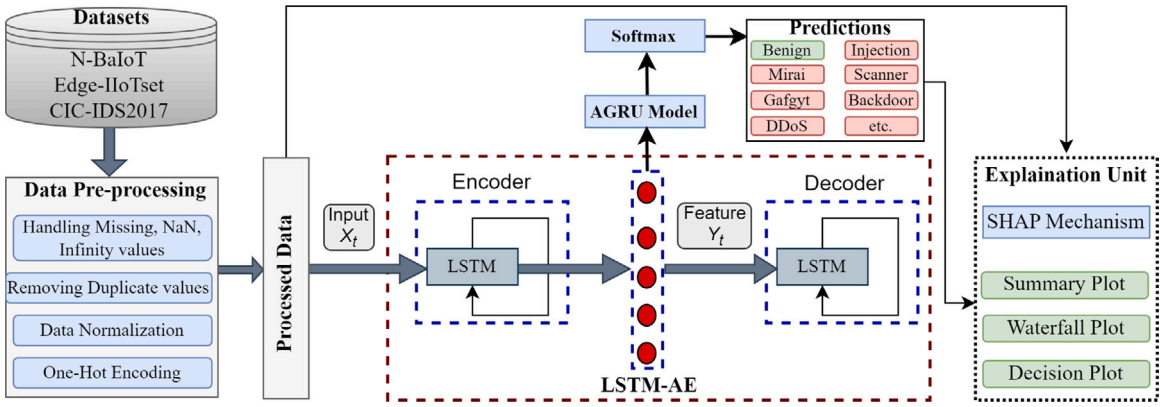


Fig. 1. Working Architecture of the Proposed Framework.

deciding what information should be transmitted to the output. They have the rare capacity to be trained to keep the information from the past without having it vanish over time. The $\mathcal{U}_t$ is calculated by Eq. (5) as follow (Dey and Salem, 2017):

$$\mathcal{U}_t = \sigma(\mathcal{W}^{(\mathcal{U})}\mathcal{D}_t + \mathcal{V}^{(\mathcal{U})}\mathcal{H}_{t-1} + \mathcal{B}_{\mathcal{U}}) \tag{5}$$

where the $\mathcal{D}_t$ represents the input, $\mathcal{W}^{(\mathcal{U})}$ is the weight, and $\mathcal{H}_{t-1}$ keeps the previous units $t-1$ information, and $\mathcal{V}^{(\mathcal{U})}$ is the weight for $t-1$ units. When $\mathcal{D}_t$ is fed to the network unit, it is multiplied by $\mathcal{W}^{(\mathcal{U})}$. Similarly, the same happens to $\mathcal{H}_{t-1}$, it is also multiplied by $\mathcal{V}^{(\mathcal{U})}$. The $\mathcal{U}_t$ is responsible for determining the amount of information from the past that needs to be passed to the future. Unlike $\mathcal{U}_t$, the $\mathcal{R}_t$ determines the amount of past information to forget. It does this by Eq. (6) (Dey and Salem, 2017).

$$\mathcal{R}_t = \sigma(\mathcal{W}^{(\mathcal{R})}\mathcal{D}_t + \mathcal{V}^{(\mathcal{R})}\mathcal{H}_{t-1} + \mathcal{B}_{\mathcal{R}}) \tag{6}$$

It uses the same EQ as $\mathcal{U}_t$. The only variation is the Weight ($\mathcal{W}$) and gate's usage. Further, it introduces a new memory content ($\hat{\mathcal{H}}_t$), which uses the rest gate $\mathcal{R}_t$ to maintain the vital past information, which is calculated as Eq. (7).

$$\hat{\mathcal{H}}_t = \tanh(\mathcal{W}\mathcal{D}_t + \mathcal{R}_t \odot \mathcal{V}\mathcal{H}_{t-1}) \tag{7}$$

The new memory content $\hat{\mathcal{H}}_t$ multiplies the input $\mathcal{D}_t$ with weight $\mathcal{W}$ and $\mathcal{H}_{t-1}$ with the $\mathcal{V}_t$. where $\odot$ represents the Hadamard product between the $\mathcal{R}_t$ and $\mathcal{V}\mathcal{H}_{t-1}$. Finally, the information of the current unit ($\mathcal{H}_t$) is required to be calculated by the network. The $\mathcal{U}_t$ plays an important role in this step as it decides what information needs to be collected from the $\hat{\mathcal{H}}_t$ and $\mathcal{H}_{t-1}$. It performs this operation through Eq. (8).

$$\mathcal{H}_t = \mathcal{U}_t \odot \mathcal{H}_{t-1} + (1 - \mathcal{U}_t) \odot \hat{\mathcal{H}}_t \tag{8}$$

We further employed an attention mechanism to make the model more efficient in threat identification. This method's primary goal is to examine threats in the system. The attention mechanism is employed as described in the following equations:

$$\mathcal{ATN}_{ij} = Softmax(\mathcal{F}_{atn}(\mathcal{H}_i, \mathcal{S}_j)) \tag{9}$$

$$\mathcal{F}_{atn}(\mathcal{H}_i, \mathcal{S}_j) = \mathcal{V}_a \sim \tanh([\mathcal{W}_a\mathcal{H}_i, \mathcal{W}_a\mathcal{S}_j]) \tag{10}$$

$$Contx_i = \sum_{\mathcal{J}} \mathcal{ATN}_{ij}\mathcal{H}_i \tag{11}$$

where $\mathcal{ATN}_{ij}$ denotes the attention scores and $\mathcal{F}_{atn}(\mathcal{H}_i, \mathcal{S}_j)$ is the attention function. Further, $\mathcal{H}_i$ is the denotes the hidden state, $\mathcal{S}_j$ represents the previous $\mathcal{H}_i$, and $Contx_i$ denotes the context vector.

3.1.4. Softmax for multiclass classification

We further employed softmax to categorize the instances of the dataset into Benign and Attack classes, such as Mirai UDP, Mirai Scan, Mirai UDP Plain, Mirai Ack, BASHLITE TCP, BASHLITE Combo, BASHLITE Junk, Ransomware, Uploading, Scanner, DDoSTCP, UDP, ICMP, and Backdoor. The equation of the softmax function is as follows (Kagalkar and Raghuram, 2020).

$$\mathcal{SX}(\overrightarrow{\mathcal{Z}_I}) = \frac{\mathcal{E}xp(\mathcal{Z}_I)}{\sum_{\mathcal{J}=1}^{\mathcal{K}} \mathcal{E}xp(\mathcal{Z}_{\mathcal{J}})} \tag{12}$$

Further, we calculated the Loss ($\mathcal{LS}$) through the Categorical-cross entropy ($\mathcal{CCE}$) $\mathcal{LS}$ function:

$$\mathcal{LS}(\hat{\mathcal{O}}_c, \mathcal{O}_c) = -\sum_{\mathcal{I}=1}^{\mathcal{N}} \sum_{c=1}^{\mathcal{C}} \mathcal{O}_c^{\mathcal{X}_i} \cdot \log(\mathcal{P}(\hat{\mathcal{O}_{Ic}} = \mathcal{O}_{Ic} \mid \mathcal{X}_i)) \tag{13}$$

where $\hat{\mathcal{O}}_c$ is the predicted and $\mathcal{O}_c$ is the actual output, $\mathcal{X}$ denotes the pattern of input sequence, $\mathcal{N}$ is the total number of observations and $\mathcal{P}$ belongs to the Output $\mathcal{O}$, such that, attack type. The working procedure of the proposed attack detection mechanism is provided in Algorithm 1.

3.1.5. Interpretability of the proposed IDSs decision

The enormous increase of DL-based IDS in Critical Infrastructures (CI) increased the demand for explaining its predictions. XAI provides the explanation of an IDS prediction in an interpretable manner that SOC analysts can easily understand. As stated by the authors of Arrieta et al. (2020), "*The new generation of artificially intelligent companions can be understood, properly trusted, and effectively controlled by human users by employing a set of AI techniques created by XAI*".. The interpretation of the model decision promotes trust and transparency. SHAP and LIME are the two most popular approaches of XAI to explain and interpret the decision of a DL-based model. These approaches learn a local interpretable model around each $\mathcal{P}$, lucidly and accurately describing the prediction made by each classifier. LIME and SHAP, estimate feature attributions on specific instances, which account for each feature's role in the black box prediction.

In this work, we employed the SHAP framework of XAI to interpret the top contributed features of the N-BaIoT, Edge-IIoTset, and CIC-IDS2017 datasets in the prediction of the proposed XDLTDS model. SHAP is introduced as a feature relevance explanation framework by the authors of Lundberg and Lee (2017) which rates the importance of each feature for each individual prediction. It provides explainability based on Missingness (MS), Local Accuracy (LA), and Consistency (CY). While using these three properties, It first recognizes a class of additive feature attribution methods, after which it shows that there is a unique solution in this class. Let D be a set of inputs and $F(D)$ is a model, SHAP's goal is to explain D's prediction by computing each feature contribution to the prediction. The authors of Lundberg and Lee (2017)

Algorithm 1 Proposed Threat Detection System

Input: N-BaIoT, Edge-IIoTset, and CIC-IDS2017 Datasets ($\mathcal{NB}, \mathcal{EI}, \mathcal{CIC}$)
Pre-process $\mathcal{NB}, \mathcal{EI}, \mathcal{CIC}$
Split $\mathcal{NB}, \mathcal{EI}, \mathcal{CIC}$ in to $\mathcal{NB}, \mathcal{EI}, \mathcal{CIC}_{Train}$ and $\mathcal{NB}, \mathcal{EI}, \mathcal{CIC}_{Test}$
Function=LSTM-AE
for $\mathcal{EPOCH}$ $i \in \{0, 1, ..., \mathrm{E}-1\}$ **do**
    Encode and Decode the data
    $\mathcal{Q}_t = \mathcal{F}(\mathcal{W}D_t + \mathcal{B})$
    $\hat{D}_t = \mathcal{F}'(\mathcal{W}'D_t + \mathcal{B}')$
    $\mathcal{M} = \frac{1}{\mathcal{P}} \sum_{i=1}^{\mathcal{P}} \mathcal{L}[D_t, \hat{D}_t]$
    $\mathcal{L}[D_t, \hat{D}_t] = ||D_t, \hat{D}_t||^2$
end for
end Function
Function = AGRU
for $\mathcal{EPOCH}$ $i \in \{0, 1, ..., \mathrm{E}-1\}$ **do**
    Train and Validate AGRU on $\mathcal{NB}, \mathcal{EI}, \mathcal{CIC}_{Train}$ encoded data using $Eq(5)$ to $Eq(11)$
    while True **do**
        Calculate AGRU training accuracy, loss, validation accuracy, validation loss.
        Calculate total loss ($\mathcal{LS}$) using $\mathcal{CC}-\mathcal{E}$ function $\mathcal{LS}(\hat{\mathcal{O}}_c, \mathcal{O}_c) = -\sum_{\mathcal{I}=1}^{\mathcal{N}} \sum_{c=1}^{C} \mathcal{O}_c^{\mathcal{X}_i} \cdot \log(\mathcal{P}(\hat{\mathcal{O}_{Ic}} = \mathcal{O}_{Ic} \mid \mathcal{X}_i))$
    end while
end for
Test the AGRU Model using $\mathcal{NB}, \mathcal{EI}, \mathcal{CIC}_{Test}$ encoded data
Generate the output using $\mathcal{SX}$;
    $\mathcal{SX}(\mathcal{Z}_\mathcal{I}) = \frac{\mathcal{E}xp(Z_\mathcal{I})}{\sum_{\mathcal{J}=1}^{\mathcal{K}} \mathcal{E}xp(\mathcal{Z}_\mathcal{J})}$
Evaluate Model using Standard Evaluation Metrics
Generate Output
end Function

represented such explanation as an Additive Feature Attribution (AFA) technique. The AFA technique defines a local input (D') simplified set that maps to the D using a mapping function $D = \mathcal{H}_D(D')$ and explanation model $\mathcal{G}(D')$:

$$\mathcal{G}(D') = \varphi_0 + \sum_{\mathcal{I}=1}^{\mathcal{N}} \varphi_\mathcal{I} D'_\mathcal{I} \tag{14}$$

where φ_0 represents the null output and $\varphi_\mathcal{I}$ denotes the feature $\mathcal{I}'s$ explained effect.

SHAP is considered to be the only post-hoc machine learning explanation model by the authors in Lundberg and Lee (2017), which satisfies the desirable properties of MS, LA, and CY. The LA property states, that input D and simplified input D' are approximately identical. The output of the $\mathcal{F}(D)$ and $\mathcal{G}(D')$ should be similar. The MS property states that if there is a missing feature from D, there will be no impact of its attribution $\varphi_\mathcal{I}$. Further, the CS property states that if $\mathcal{F}(D)$ changes, consequently, the contribution of the particular features also changes, then there should be no change to the attribution in the opposite direction. Moreover, the sampled examples are weighed by SHAP based on the weight ($\mathcal{W}$) that the simplified features would receive in the Shapely value (SV) estimate. The biggest weights are given to large simplified features (i.e., many 1's) and small simplified features (i.e., few 1's). The authors of Lundberg and Lee (2017) proposed the SHAP kernel as;

$$\pi_{\mathcal{X}'}(\mathcal{Z}') = \frac{(\mathcal{MAX} - 1)}{(\mathcal{MAX}\ choose\ |\mathcal{Z}'|)|\mathcal{Z}'|(\mathcal{MAX} - |\mathcal{Z}'|)} \tag{15}$$

where, $\mathcal{MAX}$ represents the maximum simplified features, $|\mathcal{Z}'|$ denotes the total amount of 1 entry in instance $\mathcal{Z}'$.

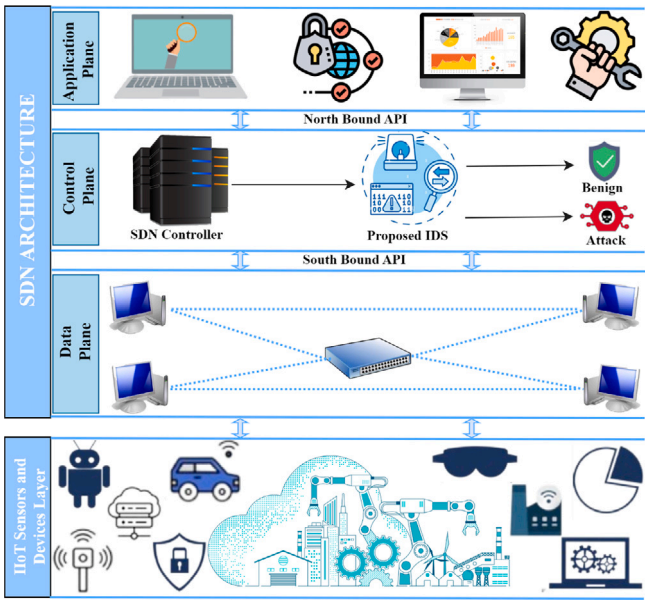


Fig. 2. Deployment architecture.

3.2. Deployment architecture

The deployment architecture for the proposed threat detection system is shown in Fig. 2. The Infrastructure Plane (IP) consists of IIoT devices, i.e., actuators, sensors, and other industrial components which are frequently used for automation and control in industrial environments and are connected to the IIoT network (Zainudin et al., 2022). These devices continuously share data with the routers or switches, placed in the Data Plane (DP) of the SDN, which subsequently communicate with the SDN controller (Zainudin et al., 2022). The controller is a part of the SDN's Control Plane (CP). The CP is considered to be the brain of SDN architecture, where the network behavior and policies are managed (Tadros et al., 2023). The data is collected from the IP and passed to the CP, where the data is processed by the SDN controller (Alshahrani et al., 2023). We integrated the proposed IDS into the CP of the SDN for the following four reasons. Firstly, SDN empowers an adaptive, dynamic, and scalable security mechanism, which means we do not need to make major architectural changes with the expansion of the network (Al Hayajneh et al., 2020). Secondly, the CP of SDN is programmable, where new rules and policies can be added to the network without having to replace any hardware (Ahmad and Mir, 2021). Thirdly, it aids SOC analysts in identifying threats efficiently by providing a statistical and event-based abstract representation of the network that shows what is happening in the network, which results in low false alarm rates and minimizes unnecessary alerts. Traffic is categorized as benign or malicious after being analyzed by the proposed IDS. Consequently, it reduces the potential damage to the network. Fourthly, deployment of the proposed TDS in SDN optimizes resource usage and does not overburden the resource-constrained devices, thus ensuring the overall reliability and security of the IIoT networks. Finally, the Application Plane (AP), which is the highest layer, is where different network services and applications communicate with the SDN controller, either directly or through southbound and northbound APIs. This layer provides numerous applications such as monitoring, optimization analytics, and security (Shaghaghi et al., 2020). Once the decision is made by the proposed IDS in the CP, it is sent to the AP of the SDN where the SOC analysts are monitoring the network. Consequently, it helps them identify the threats efficiently and relieves them from heavy investigation tasks.

Table 2
Implementation details.

Parameters of LSTM-AE	
Details	Hyperparameters
Input Layer	115 features of the N-BaIoT dataset 46 features of the Edge-IIoTset dataset 78 features of the CIC-IDS2017 dataset
Encoding Layers	First Layer: 80 units, ReLU activation Second Layer: 50 units, ReLU activation
Decoding Layers	Repeat Vector: Time steps: original sequence length LSTM Layer: 50 units, ReLU activation, return sequences: True LSTM Layer: 80 units, ReLU activation, return sequences: True
Loss Function Optimizer Epochs Batch-size	Mean Square Error (MSE) Adam 10 64
Parameters of AGRU Model	
Details	Hyperparameters
Input Layer	Encoded features from the LSTM-AE
GRU Layers	First Layer: 80 units, ReLU activation, return sequences: True Second Layer: 50 units, ReLU activation, return sequences: True
Dropout	Rate: 0.2
Attention Layer	Weight matrix shape: (input_dim, 1), Bias shape: (time_steps, 1)
Output Layer	08 units (01 Benign and 07 attack classes for N-BaIoT), 08 units (01 Benign and 07 attack classes for Edge-IIoTset), 11 units (01 BENIGN and 10 attack classes for CIC-IDS2017), Softmax function
Loss Function Optimizer Epochs Batch-size	Categorical Crossentropy (CC-E) Adam 10 64

4. Experimental setup & Evaluation metrics

4.1. Implementation setup

The experiments were conducted using the Keras framework on the TensorFlow platform with Python version 3.8. We utilized the Spyder IDE for experimentation. For comprehensive details on the architecture and hyperparameters of the proposed model, please refer to Table 2.

4.2. Datasets description

To evaluate our proposed framework, we employed three publicly available datasets, namely N-BaIoT (Meidan et al., 2018), Edge-IIoTset (Ferrag et al., 2022), and CIC-IDS2017 (Sharafaldin et al., 2018) datasets. We chose these datasets because they represent the IIoT network traffic well (especially edge traffic). The N-BaIoT dataset comprises a benign class with two types of attack categories, such as Mirai and Gafgyt, which have various subtypes. The Edge-IIoTset dataset also comprises a normal class and different attack classes. Further, CIC-IDS2017 has one BENIGN class and more than ten attack classes. In this work, we only consider eight classes of the N-BaIoT and Edge-IIoTset datasets to evaluate our proposed model as these attacks are the most frequent in industries. Regarding the CIC-IDS2017 dataset, we have considered all the classes except infiltration and heartbleed classes, as they have very few samples, i.e., 36 and 11. Furthermore, the selected classes of the N-BaIoT dataset include Benign, Mirai_udp, Mirai_syn, Mirai_ack, Mirai_scan, Gafgyt_combo, Gafgyt_junk, and Gafgyt_scan. Regarding the EdgeIIoTset, the selected classes are Normal, DDoS_UDP, DDoS_ICMP, SQL_injection, DDoS_TCP, Vulnerability_scanner, DDoS_HTTP, and Backdoor. The complete details about the instances of each class used for training and testing in this work are provided in Table 3.

4.2.1. Pre-processing

The data in the datasets are usually unstructured and in raw form. It requires preprocessing before taking it to train the model, which makes it usable and understandable by the machine. We first imputed all the Null values using the Mean. Further, we removed all the rows containing NaN values, followed by removing duplicate values. We used the LabelEncoder to convert the categorical variables to numerics. Moreover, we used the Minmax Scaler function of sklearn library to normalize the data. The normalization is done by Eq. (16):

$$\mathcal{Y}_S = \frac{\mathcal{Y} - \mathcal{Y}_{\mathcal{MIN}}}{\mathcal{Y} - \mathcal{Y}_{\mathcal{MAX}}} \tag{16}$$

where, $\mathcal{Y}_S$ is the new value and $\mathcal{Y}$ denotes the value of original cell. Further, $\mathcal{Y}_{\mathcal{MIN}}$ is the minimum value of the column which is 0, and $\mathcal{Y}_{\mathcal{MAX}}$ represents the maximum value which is 1. In other words, all the values in the data will now be between 0 and 1. We further employed One-hot encoding to the output labels to transform them into numerical values. Finally, we divided the N-BaIoT and Edge-IIoTset datasets into 70% and 30% ratios for training and testing purposes.

4.3. Performance evaluation metrics

In this work, several metrics of evaluation have been employed to evaluate the proposed XDLTDS's performance. The details of these metrics are as follows (Javeed et al., 2023c);

4.3.1. Accuracy (ACC)

It is a measurement of how accurately a classification model predicts or categorizes data items. Eq. (17) is used to calculate this metric.

$$ACC = \frac{TPR + TNR}{TPR + TNR + FPR + FNR} \tag{17}$$

Table 3
Datasets instances detail.

Dataset	Class	Total instances	Training instances	Testing instances
N-BaIoT	Benign	201234	140613	60621
	Mirai_udp	137689	96212	41477
	Mirai_syn	82207	57665	24542
	Mirai_ack	70740	49621	21119
	Mirai_scan	57398	40318	17080
	Gafgyt_combo	33270	23325	9945
	Gafgyt_junk	17265	12111	5154
	Gafgyt_scan	16549	11581	4968
Edge-IIoTset	Normal	1363998	954810	409188
	DDoS_UDP	121567	85083	36484
	DDoS_ICMP	67939	47541	20398
	SQL_injection	50826	35508	15318
	DDoS_TCP	50062	35167	14988
	Vulnerability_scanner	50026	35038	14895
	DDoS_HTTP	48544	34012	14532
	Backdoor	24026	16732	7294
CIC-IDS2017	BENIGN	2095057	1466519	628538
	DoS Hulk	172846	121111	51735
	DDoS	128014	89689	38325
	PortScan	90694	63299	27395
	DoS GoldenEye	10286	7206	3080
	FTP-Patator	5931	4131	1800
	DoS slowloris	5385	3763	1622
	DoS Slowhttptest	5228	3682	1546
	SSH-Patator	3219	2263	956
	Web Attack	2143	1472	671
	Bot	1948	1390	558

4.3.2. Recall (REC)

By comparing the proportion of TP to all other positive instances in the dataset, it assesses a model's capacity to find all pertinent occurrences within a dataset. Eq. (18) performs this operation.

$$REC = \frac{TPR}{TPR + FNR} \tag{18}$$

4.3.3. Precision (PRE)

It is the proportion of accurate positive predictions to all of the model's positive predictions. Eq. (19) is used to calculate the PRE.

$$PRE = \frac{TPR}{TPR + FPR} \tag{19}$$

4.3.4. F1-score (F1)

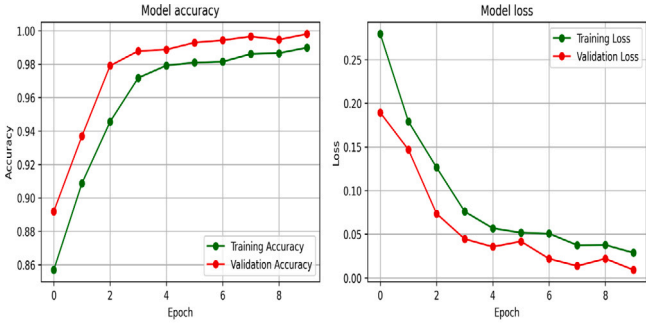
It considers PRE and REC at the same time to demonstrate the ACC of a classification model in a balanced way. Eq. (20) performs such an operation.

$$F1 = 2 \times \frac{PRE \times REC}{PRE + REC} \tag{20}$$

where TPR is the proportion of actual positives that are accurately detected by a model and FPR is the proportion of actual negatives that are incorrectly classified as positives. Further, TNR is the proportion of actual negatives accurately detected by a model, While FNR is the proportion of actual positives that are incorrectly classified as negatives.

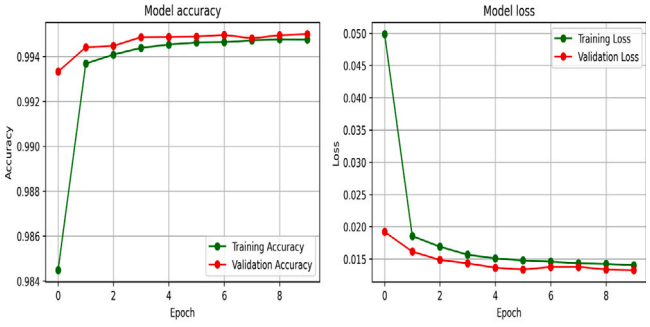
5. Results & Analysis

In this section, we discuss the results and evaluate the performance of the proposed XDLTDS framework. First, we present and analyze its performance using standard evaluation metrics, followed by an assessment of the real-time detection feasibility of the proposed IDS. Additionally, we compare its performance with state-of-the-art mechanisms from the existing literature.



(a) Accuracy (b) Loss

Fig. 3. Acc vs. Loss using N-BaIoT dataset.



(a) Accuracy (b) Loss

Fig. 4. Acc vs. Loss using Edge-IIoTset dataset.

5.1. Proposed scheme performance analysis

We conducted a thorough evaluation of the performance of the proposed XDLTDS using the N-BaIoT, Edge-IIoTset, and CIC-IDS2017 datasets. Fig. 3 shows the model's performance during training and

Table 4
Proposed model performance on N-BaIoT dataset.

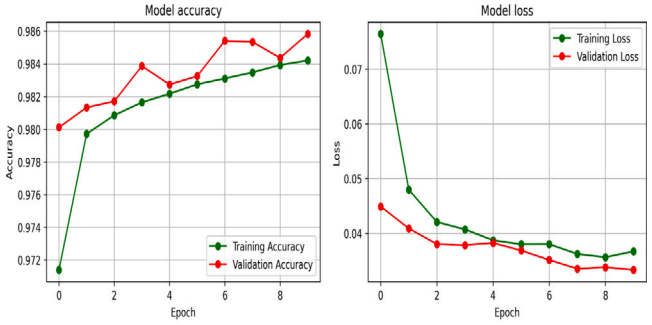
Evaluation metrics	Benign	Mirai_udp	Mirai_syn	Mirai_ack	Mirai_scan	Gafgyt_combo	Gafgyt_junk	Gafgyt_scan
PRE	99.97%	99.98%	100.00%	99.85%	100.00%	99.55%	95.81%	98.90%
REC	99.93%	99.91%	99.95%	99.95%	99.94%	97.77%	99.12%	99.89%
F1	99.95%	99.95%	99.97%	99.90%	99.97%	98.66%	97.44%	99.39%
FPR	0.0001126	0.0000418	0.00	0.0001892	0.00	0.0002457	0.0012405	0.0003056
Accuracy					99.79%			
Macro average PRE					99.26%			
Macro average REC					99.56%			
Macro average F1					99.40%			

Table 5
Proposed model performance on Edge-IIoTset dataset.

Evaluation Metrics	Normal	DDoS_UDP	DDoS_ICMP	SQL_injection	DDoS_TCP	Vulnerability_scanner	DDoS_HTTP	Backdoor
PRE	99.99%	99.94%	99.99%	99.98%	98.74%	98.34%	86.38%	99.98%
REC	100.00%	100.00%	99.90%	99.98%	100.00%	84.94%	98.51%	97.36%
F1	99.99%	99.97%	99.94%	99.98%	99.36%	91.15%	92.05%	98.65%
FPR	0.0000161	0.0000402	0.0000039	0.0000038	0.0003647	0.0004130	0.0043504	0.0000019
Accuracy					99.49%			
Macro average PRE					97.92%			
Macro average REC					97.58%			
Macro average F1					97.64%			

Table 6
Proposed model performance on CIC-IDS2017 dataset.

Evaluation Metrics	BENIGN	DoS Hulk	DDoS	PortScan	DoS GoldenEye	FTP-Patator	DoS slowloris	DoS Slowhttptest	SSH-Patator	Web Attack	Bot
PRE	99.32%	97.49%	99.83%	84.29%	97.35%	99.32%	89.32%	87.78%	99.88%	95.08%	96.69%
REC	98.98%	98.21%	99.86%	92.13%	98.08%	98.77%	98.52%	91.59%	91.73%	08.64%	36.73%
F1	99.15%	97.85%	99.85%	88.04%	97.71%	99.05%	93.69%	89.64%	95.63%	15.84%	53.24%
FPR	0.033166	0.001856	0.000086	0.006451	0.000108	0.000015	0.000253	0.000261	0.000001	0.000003	0.000009
Accuracy						98.57%					
Macro average PRE						95.12%					
Macro average REC						83.02%					
Macro average F1						84.52%					



(a) Accuracy (b) Loss

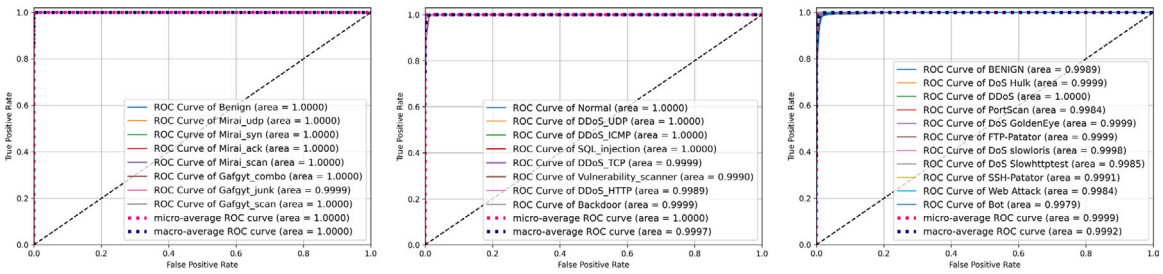
Fig. 5. Acc vs. Loss using CIC-IDS2017 dataset.

validation for the N-BaIoT dataset. Initially, the accuracy and loss of the model struggled during the first four epochs, but in later epochs, the model's performance improved significantly, as shown in Figs. 3(a) and 3(b) for accuracy and loss, respectively. For the Edge-IIoTset dataset, the model exhibited stable performance from the beginning, provided in Fig. 4. Although the training accuracy and loss were slightly lower in the first epoch compared to the validation metrics, the model quickly achieved similar performance in subsequent epochs, as shown in Figs. 4(a) and 4(b). Further, the training and validation performance of the model using the CIC-IDS2017 dataset is provided in 5(a) and 5(b). The test accuracy of the model for all the datasets is provided in Tables 4, 5, and 6 respectively.

Furthermore, ROC curve assesses the performance of a DL-based IDS by graphical representation. It is created by plotting the TPR against the FPR at various threshold settings for the model. The curve serves as a visual depiction of how effectively the model can discriminate between the attack classes, with each point on the curve denoting a different threshold. A typical statistic for assessing the general effectiveness of a threat detection model is the AUC of the ROC. A higher value of AUC-ROC, i.e., close to 1 indicates the model efficiency in threat detection. Fig. 6(a) depicts the ROC curve of the proposed XDLTDS framework for the N-BaIoT dataset, Fig. 6(b) depicts the ROC curve for the Edge-IIoTset dataset, while Fig. 6(c) for the CICIDS-2017 dataset. The AUC-ROC values for all the classes are almost equal to 1 as shown in the figures. Further, the micro average and macro average achieved by the model are both 1.00 under the N-BaIoT dataset, 1.00 and 0.9997 under the Edge-IIoTset dataset, and 0.9999 and 0.992 under the CIC-IDS2017 dataset, which demonstrates the efficient performance of our proposed framework in detecting the normal class and attack classes.

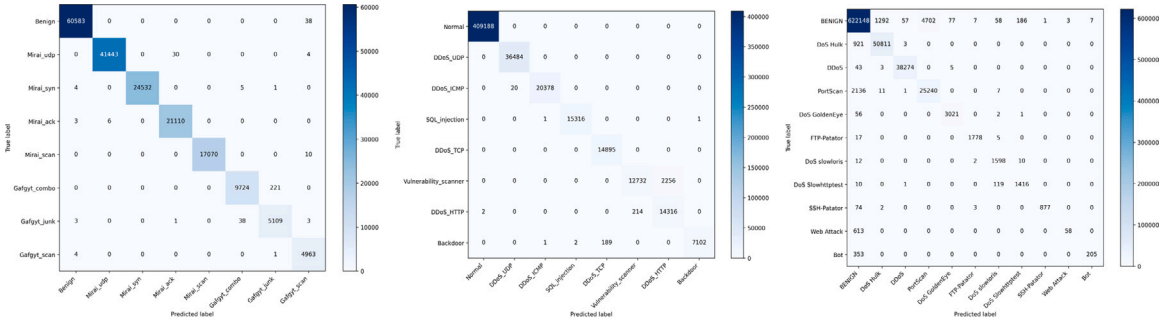
Confusion Matrix (CM) summarizes the performance of DL-based IDS on test data. It is used to evaluate how well a model classifies the attack classes by predicting a category label for each input event. It displays the number of TP, FP, TN, and FN produced by the threat detection model on the test data. It provides a summary of the number of records the threat detection scheme properly or erroneously detected. In a CM, a row corresponds to an instance of a certain class, while a column indicates it of a predicted class. If there are fewer instances off the diagonal in a CM, then the classification model is considered ideal. The CM of the proposed framework is provided in Fig. 7, where Fig. 7(a) depicts the CM under the N-BaIoT dataset, Fig. 7(b) presents the CM under the Edge-IIoTset dataset, and Fig. 7(c) under the CIC-IDS2017 dataset.

Furthermore, we provide a complete evaluation of the proposed XDLTDS framework by considering standard evaluation metrics. The values for these metrics are computed by using Eq. (17) to Eq. (20) discussed in Section 4-B. We provide the class-wise performance of the proposed XDLTDS framework in terms of PRE, REC, F1, and FPR. Table 4 presents the class-wise performance for the N-BaIoT dataset, while Table 5 presents the class-wise performance for the



(a) AU-ROC values on N-BaIoT Dataset (b) AU-ROC values on Edge-IIoT Dataset (c) AU-ROC values on CIC-IDS2017 Dataset

Fig. 6. In these ROC Curves, each class from the datasets is evaluated based on the FPR and TPR depicted on the X-axis and Y-axis respectively.



(a) Using N-BaIoT Dataset. (b) Using Edge-IIoTset Dataset. (c) Using CIC-IDS2017 Dataset.

Fig. 7. These CMs provide the evaluation of the XDLTDS framework classification performance of different classes. The predicted labels are presented on the X-axis, while the true labels are on the Y-axis.

Table 7
Comparison with state-of-the-art Mechanisms.

Dataset	Authors	Year	Framework	ACC	REC	PRE	F1
N-BaIoT	Taher et al. (2023)	2023	IHHO-NN	98.07%	98.73%	97.04%	97.87%
	Santhadevi and Janet (2023)	2023	CNN-SLSTM	97.39%	97.83%	95.77%	96.79%
	Proposed	2024	XDLTDS	**99.79%**	**99.56%**	**99.26%**	**99.40%**
Edge-IIoTset	Ferrag et al. (2022)	2022	DNN	96.01%	–	–	–
	Tareq et al. (2022)	2022	Inception Time	94.94%	92.40%	**98.30%**	95.30%
	Ding et al. (2023)	2023	DeepAK-IoT	94.96%	–	95.40%	94.72%
	Friha et al. (2023)	2023	2DF-IDS	94.37%	91.00%	88.00%	89.00%
	Proposed	2024	XDLTDS	**99.49%**	**97.58%**	97.92%	**97.64%**
CIC-IDS2017	Zhang et al. (2024)	2024	HPPELM	94.67%	82.25%	89.59%	**85.83%**
	Proposed	2024	XDLTDS	**98.57%**	**83.02%**	**95.12%**	84.52%

Terms & Abbreviations: 2DF-IDS: Decentralized Differentially Private Intrusion Detection System; DA: Deployment Architecture; RTD Analysis: Real-time Detection Analysis; CNN: Convolutional Neural Network; SLSTM: Stacked Long short-term Memory; IHHO-NN: Improved Harris Hawks Optimization Algorithm for Neural Networks.

Edge-IIoTset dataset. Further, the class-wise performance for the CIC-IDS2017 dataset is provided in Table 6. The tables show that the proposed model has achieved high PRE, REC, and F1 for all the classes of the datasets with a very low false positive rate. Moreover, we also provide the average results of the proposed XDLTDS framework in Table 4, Table 5, and Table 6. The proposed framework achieved ACC of 99.79% with macro average PRE, REC, and F1 of 99.26%, 99.56%, and 99.40% using the N-BaIoT dataset as shown in Table 4. Furthermore, the proposed framework achieved ACC and macro average PRE of 99.49% and 97.92% on the Edge-IIoTset dataset. While for REC and F1, it has values of 97.58%, and 97.64% respectively. Moreover, it has 98.57% ACC, with macro average PRE, REC, and F1 of 95.12%, 83.02%, and 84.52% using the CIC-IDS2017 dataset. The reason for achieving low REC and F1 using the CIC-IDS2017 dataset is the smaller number of samples for Web Attack and Bot classes in the dataset.

5.2. Real-time detection feasibility

The effectiveness of an IDS depends not only on its overall performance but also on its ability to detect attacks promptly. An IDS must identify potential threats in real-time; otherwise, delayed detection renders it ineffective if the attack has already occurred. To assess the real-time detection capability of the proposed IDS, we analyzed its detection times across all datasets in a simulated environment. Our proposed IDS demonstrated impressive responsiveness, with detection times of 0.079 ms for the N-BaIoT dataset, 0.087 ms for the Edge-IIoTset, and 0.084 ms for the CIC-IDS2017 dataset. These results were obtained using a moderate hardware configuration: Intel(R) Core(TM) i5-6200U CPU @ 2.30 GHz with 12 GB of RAM, reflecting the resource constraints commonly encountered in IoT network environments. The time detection comparison of the proposed IDS for different datasets is provided in Fig. 8.

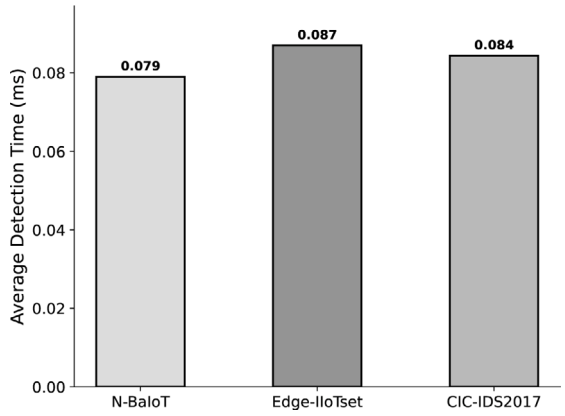


Fig. 8. Proposed IDS detection time per sample.

5.3. Proposed scheme comparison with state-of-the-art mechanisms

This subsection presents a performance comparison between our proposed XDLTDS scheme and several of the most recent threat detection techniques from the existing literature, i.e., Taher et al. (2023), Santhadevi and Janet (2023), Ferrag et al. (2022), Tareq et al. (2022), Ding et al. (2023), Friha et al. (2023), and Zhang et al. (2024). The authors of Tareq et al. (2022) achieved a detection ACC of 94.94%, REC of 92.40%, PRE of 98.30% and F1 of 95.30%, while (Ding et al., 2023) has ACC of 94.96% with PRE and F1 of 95.40% and 94.72%. The authors did not provide the REC in their work. Further, the authors of Friha et al. (2023) have achieved an ACC of 94.37%, REC of 91.00%, PRE of 88.00%, and F1 of 89.00%. Moreover, Taher et al. (2023) achieved ACC of 98.07%, REC of 98.73%, PRE of 97.04%, and F1 of 97.87% respectively. Additionally, the authors of Santhadevi and Janet (2023) achieved ACC of 97.39%, REC of 97.83%, PRE of 95.77%, and F1 of 96.79%. Finally, the authors of Zhang et al. (2024) achieved ACC OF 94.67% with REC, PRE, and F1 of 82.25%, 89.59%, and 85.83% respectively. The complete comparison is provided in Table 7. The highest values are shown in bold. It can be seen that the proposed model outperformed the existing techniques in terms of all the evaluation metrics by achieving higher values, except the authors of Tareq et al. (2022), with 0.38% higher precision under Edge-IIoTset dataset and the authors of Zhang et al. (2024), with 1.31% higher F1 under the CIC-IDS2017 dataset, thus proving its efficient performance.

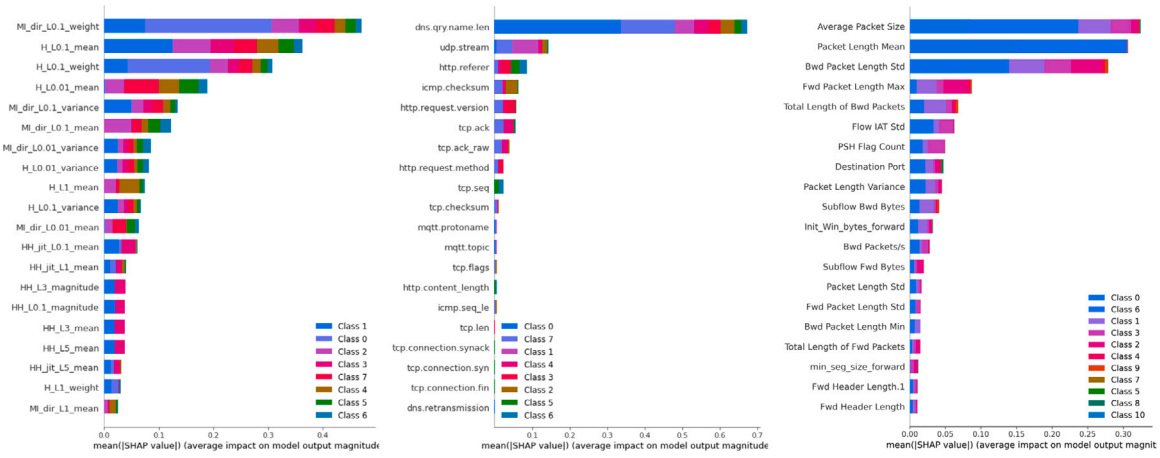
Consequently, this subsection demonstrates that the proposed XDLTDS framework has shown a good performance and confirms that it is a promising solution for industries to address the challenges of modern cybersecurity. It can effectively defend networks against threats, ensuring network security and privacy.

6. XAI-based interpretation

This section presents the interpretability of the proposed XDLTDS mechanism by explaining the learned features of the data that contribute the most to the model's decision. We employed several plots provided by XAI to interpret the features' contributions to the predictions of the proposed XDLTDS framework, enhancing trust and transparency.

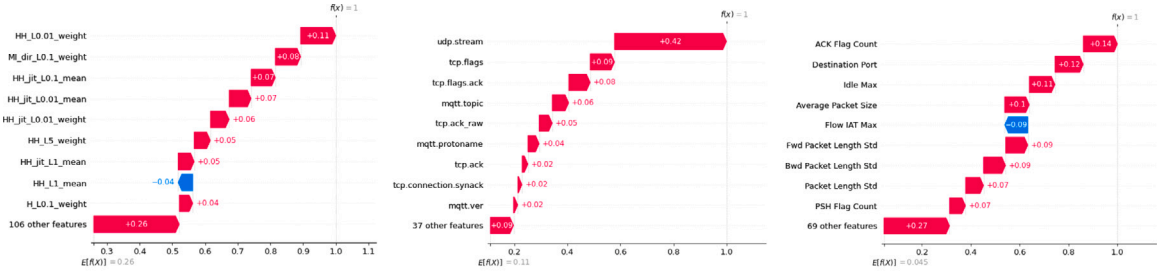
The Summary (SM) plot combines the significance of the features with their impacts. Each point on the SM plot represents an SV for an instance and a feature. The feature determines the position on the y-axis, while the SV determines the position on the x-axis. Fig. 9 depicts the SM plot of the proposed XDLTDS framework that provides the global feature importance values that are obtained from the training data. The X-axis of Fig. 9(a) shows the average impact of the features of the N-BaIoT dataset on model's prediction. Further, Fig. 9(b) shows the impact of the feature of the Edge-IIoTset dataset on the model prediction. Fig. 9(c) shows the impact of the features of the CIC-IDS2017 dataset on model's prediction. The features from the complete training sets are displayed on the Y-axis of the figure in order of their overall relevance. Fig. 9(a) shows that Ml_dir_L0.1_weight, H_L0.1_mean, H_L0.1_weight, H_L0.01_mean, Ml_dir_L0.1_variance, Ml_dir_L0.01_mean, mL_dir_L0.01_variance, H_L0.01, etc, features have the most contribution to the model decision. Further, Fig. 9(b) shows that the dns.qry.name.len, udp.stream, http.refere, icmmp.checksum, http.request.version, tcp.ack, tcp. ack_raw, and http.request.method features of the Edge-IIoTset dataset contributed heavily to the decision of the proposed detection scheme in identifying numerous threats. Further, Fig. 9(c) shows that Average Packet Size, Packet Length Mean, Bwd Packet Length Std, Fwd Packet Length Max, Total Length of Bwd Packets, and Flow IAT Std, etc have the most contribution to the model decision.

Further, the Waterfall (WF) plot is specifically designed to illustrate explanations for single-instance problems. The WF plot takes a single row of an explanation object as input. The Expected Value (EV) of the model output is shown at the bottom of the WF plot. In such a figure, each feature's contribution is depicted for the selected instance, with blue denoting negative contributions and red representing positive ones. We have selected random instances to analyze how the model predicts and what contributions each feature makes to that prediction. This allows us to visualize the impact of each feature on the model's decision for a specific instance. Specifically, we randomly selected the sixteenth instance of the Mirai_udp class from the N-BaIoT dataset, the twenty-fifth instance of the DDoS_UDP class of Edge-IIoTset dataset, and the tenth instance of the CIC-IDS2017 dataset to check the contribution of the features in the detection of these instances. Fig. 10(a) shows the WF plot for the N-BaIoT dataset, Fig. 10(b) shows the WF plot for the Edge-IIoTset dataset, and Fig. 10(c) shows the WF plot for the CIC-IDS2017 dataset, illustrating each feature's contribution to the proposed model's decision. It can be seen that all the features have positive contributions to the model output except the HH_L1_mean feature of the N-BaIoT dataset and the Flow IAT Max feature of the CIC-IDS2017 dataset contributed negatively to the detection of the given instances. Finally, the Decision (DN) plot is considered to be very effective for tracing the decision-making process of a classification model. It shows how the value of each of the features influences the overall decision of a model for an individual instance. For the DN plot, we have selected the same instances that we did for the WF plot, i.e., the sixteenth instance of the Mirai_udp class from the N-BaIoT dataset, the twenty-fifth instance of the DDoS_UDP class from Edge-IIoTset dataset, and the tenth instance of the DoS Hulk class from the CIC-IDS2017 dataset. The DN plots of the N-BaIoT, Edge-IIoTset, and CIC-IDS2017 datasets are provided in Figs. 11(a), 11(b), and 11(c) respectively. The colored line in these figures is the prediction, while the values next to the prediction line are the specific numerical values that each feature takes on for the given instance being evaluated. These feature values indicate the magnitude or presence of each characteristic that the model considers when making its prediction. The features that push the prediction to the right increase the model output, while the features that pull the prediction to the left decrease the model output. For example, for the HH_L1_mean feature in Fig. 11(a), the number 0.102 indicates that, for this particular data point, the values of the HH_L1_mean feature were 0.102. In Fig. 11(b), the values for tcp.flags, tcp.flags.ack, mqtt.topic, and tcp.ack_raw were 0.96, 1, 0.5, and 0.814. Further, in Fig. 11(c), the value of the features was 0.001 for Destination Port, 0.409 for Idle Max, 0.02 for Average Packet Size, and 0.409 for Flow IAT Max features. The values are shown for the top 20 contributed features of the dataset in the model output. The red line in Fig. 11(a) primarily moves to the right, indicating that all the key features contribute positively to the final model output except for the HH_L1_mean, H_L0.01_Weight, and HH_L1_magnitude feature, they



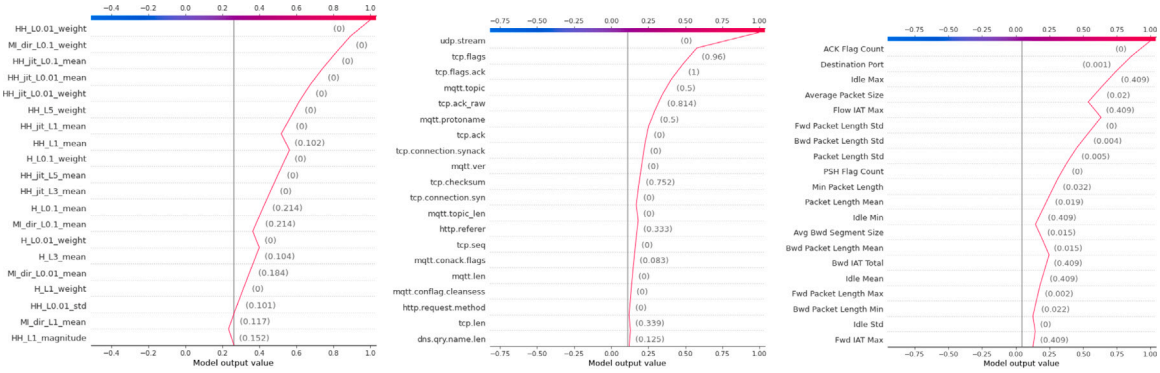
(a) Using N-BaIoT Dataset (b) Using Edge-IIoTset Dataset (c) Using CIC-IDS2017 Dataset

Fig. 9. Global explanation using SM plot.



(a) Using N-BaIoT Dataset (b) Using Edge-IIoTset Dataset (c) Using CIC-IDS2017 Dataset

Fig. 10. Local explanation using WF plot.



(a) Using N-BaIoT Dataset (b) Using Edge-IIoTset Dataset (c) Using CIC-IDS2017 Dataset

Fig. 11. Local explanation using DN plot.

have negative contributions to the model output as they are pushing the line towards the left. This visualization underscores the model's interpretability, showing how each feature in the detection of a single instance shifts the prediction towards or away from the final outcome. Such insights are valuable in identifying the driving factors behind the model's decisions, improving both transparency and trustworthiness in DL-based IDSs.

7. Discussion

Detecting sophisticated threats in IIoT networks is a challenging task due to its heterogeneous and resource-constrained nature, where each device follows a different protocol and requires a different security mechanism. In the last few years, a variety of mechanisms has been proposed by the researchers. However, they encounter significant challenges while designing an efficient IDS for such a network due to the heterogeneous and resource-constrained nature of IIoT devices. These challenges include achieving a promising accuracy and detection rate, interpretation of their predictions to promote trust, and deployment architecture of the proposed IDS to provide solutions for the resource-constrained nature of devices in the IIoT network. Consequently, these challenges make them impractical to deploy for critical networks like IIoT. To address the aforementioned challenges, we propose an XAI-integrated DL-based framework for resilient and transparent threat detection in IIoT-based networks.

XAI-integrated IDS offers a multifaceted improvement to the security operations of IIoT networks. It makes the workings of the IDS transparent by interpreting the features of the data and explaining

the decision of the IDS in understandable terms as to why a specific network traffic is flagged as a threat by the IDS. Such an explanation of the decision is of paramount importance as it allows security operation center analysts to quickly take action against such threats and precisely address the potential vulnerabilities in the network, thereby bolstering the system's defense mechanisms. Additionally, this level of explainability and transparency of IDS promotes trust between the stakeholders, who might otherwise be wary of relying solely on the impervious decisions of a black box IDS. Moreover, XAI-integrated IDS assists in meeting strict regulatory demands frequently obligatory in critical infrastructure, by offering a clear audit trail of the IDS's decision-making process. Ultimately, XAI-integrated IDS in industrial settings not only improves security but also integrates more seamlessly with human oversight, ensuring that defensive measures are both robust and understandable. This integration of XAI and DL improves overall operational efficiency, making security protocols more proactive and informed. Consequently, it allows industries to defend against sophisticated threats efficiently and reduces the risk of sensitive data loss and reputation loss.

The proposed framework consists of three modules: (1) an LSTM-AE to encode the data of the IIoT network for preventing inference attacks, (2) an attention-based GRU combined with softmax for threat detection, (3) an XAI-based technique to interpret its decision to promote trust and transparency. Further, we provide a deployment architecture for empowering a dynamic, adaptive, and scalable security mechanism. The framework is trained and evaluated on three state-of-the-art datasets, i.e., N-BaIoT, Edge-IIoTset, and CIC-IDS2017 datasets containing a comprehensive package of diversified relevant attacking instances for IIoT environments. To provide an in-depth assessment of the proposed framework's performance, we compare its performance with existing state-of-the-art mechanisms in Table 7. The proposed model outclasses the recent state-of-the-art mechanisms.

8. Conclusion

In this paper, we develop the "XDLTDS" threat detection system to address the growing cybersecurity challenges in industrial networks. Our system leverages the combined strengths of XAI and DL techniques. The proposed system employs LSTM-AE to encode IIoT data, mitigating the risk of inference attacks, and uses the AGRU model with softmax for multiclass classification of cyber threats. To enhance transparency, we integrate the SHAP mechanism, which assists cybersecurity analysts in understanding the detection process. Additionally, the inclusion of SDN provides a flexible and scalable deployment architecture. Extensive simulations on benchmark datasets validate XDLTDS's superior performance in identifying modern cyber threats in IIoT communications, outperforming existing frameworks. In future work, we plan to expand XDLTDS into a privacy-preserving collaborative framework to safeguard industrial networks across dispersed geographical locations.

CRediT authorship contribution statement

Shifa Shoukat: Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Tianhan Gao:** Writing – review & editing, Supervision, Formal analysis. **Danish Javeed:** Writing – original draft, Validation, Formal analysis, Conceptualization. **Muhammad Shahid Saeed:** Writing – review & editing, Writing – original draft. **Muhammad Adil:** Writing – review & editing, Validation, Formal analysis.

Declaration of competing interest

The authors declare that they have no significant competing financial, professional, or personal interests that might have influenced the performance or presentation of the work described in this manuscript.

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Data availability

Data will be made available on request.

References

Adil, M., Jan, M.A., Liu, Y., Abulkasim, H., Farouk, A., Song, H., 2022. A systematic survey: Security threats to UAV-aided IoT applications, taxonomy, current challenges and requirements with future research directions. IEEE Trans. Intell. Transp. Syst. 24 (2), 1437–1455.

Ahmad, S., Mir, A.H., 2021. Scalability, consistency, reliability and security in SDN controllers: a survey of diverse SDN controllers. J. Netw. Syst. Manage. 29, 1–59.

Ahmed, I., Jeon, G., Piccialli, F., 2022. From artificial intelligence to explainable artificial intelligence in industry 4.0: a survey on what, how, and where. IEEE Trans. Ind. Inform. 18 (8), 5031–5042.

Al Hayajneh, A., Bhuiyan, M.Z.A., McAndrew, I., 2020. Improving internet of things (IoT) security with software-defined networking (SDN). Comput. 9 (1), 8.

Alshahrani, H., Khan, A., Rizwan, M., Reshan, M.S.A., Sulaiman, A., Shaikh, A., 2023. Intrusion detection framework for industrial internet of things using software defined network. Sustain. 15 (11), 9001.

Arrieta, A.B., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., García, S., Gil-López, S., Molina, D., Benjamins, R., et al., 2020. Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. Inf. Fusion 58, 82–115.

Arthi, R., Krishnaveni, S., Zeadally, S., 2024. An intelligent SDN-IoT enabled intrusion detection system for healthcare systems using a hybrid deep learning and machine learning approach. China Commun..

Ashraf, I., Park, Y., Hur, S., Kim, S.W., Alroobaea, R., Zikria, Y.B., Nosheen, S., 2022. A survey on cyber security threats in iot-enabled maritime industry. IEEE Trans. Intell. Transp. Syst. 24 (2), 2677–2690.

Chen, D., Zhao, Z., Qin, X., Luo, Y., Cao, M., Xu, H., Liu, A., 2020. MAGLeak: A learning-based side-channel attack for password recognition with multiple sensors in IIoT environment. IEEE Trans. Ind. Inform. 18 (1), 467–476.

Chiu, W.-Y., Meng, W., Ge, C., 2022. NoSneaky: A blockchain-based execution integrity protection scheme in industry 4.0. IEEE Trans. Ind. Inform..

Dey, R., Salem, F.M., 2017. Gate-variants of gated recurrent unit (GRU) neural networks. In: 2017 IEEE 60th International Midwest Symposium on Circuits and Systems. MWSCAS, IEEE, pp. 1597–1600.

Ding, W., Abdel-Basset, M., Mohamed, R., 2023. DeepAK-IoT: An effective deep learning model for cyberattack detection in IoT networks. Inform. Sci. 634, 157–171.

Ferrag, M.A., Friha, O., Hamouda, D., Maglaras, L., Janicke, H., 2022. Edge-IIoTset: A new comprehensive realistic cyber security dataset of IoT and IIoT applications for centralized and federated learning. IEEE Access 10, 40281–40306.

Friha, O., Ferrag, M.A., Benbouzid, M., Berghout, T., Kantarci, B., Choo, K.-K.R., 2023. 2DF-IDS: Decentralized and differentially private federated learning-based intrusion detection system for industrial IoT. Comput. Secur. 127, 103097.

Hasan, M.K., Sulaiman, R., Islam, S., Rehman, A.U., et al., 2023. An explainable ensemble deep learning approach for intrusion detection in industrial internet of things. IEEE Access.

Illy, P., Kaddoum, G., de Araujo-Filho, P.F., Kaur, K., Garg, S., 2022. A hybrid multistage DNN-based collaborative IDPS for high-risk smart factory networks. IEEE Trans. Netw. Serv. Manag. 19 (4), 4273–4283.

Javeed, D., Gao, T., Saeed, M.S., Khan, M.T., 2023a. FOG-empowered augmented intelligence-based proactive defensive mechanism for IoT-enabled smart industries. IEEE Internet Things J..

Javeed, D., Gao, T., Saeed, M.S., Kumar, P., 2023b. An intrusion detection system for edge-envisioned smart agriculture in extreme environment. IEEE Internet Things J..

Javeed, D., Gao, T., Saeed, M.S., Kumar, P., Kumar, R., Jolfaei, A., 2023c. A softwarized intrusion detection system for iot-enabled smart healthcare system. ACM Trans. Internet Technol..

Kagalkar, A., Raghuram, S., 2020. CORDIC based implementation of the softmax activation function. In: 2020 24th International Symposium on VLSI Design and Test. VDAT, IEEE, pp. 1–4.

Keshk, M., Koroniotis, N., Pham, N., Moustafa, N., Turnbull, B., Zomaya, A.Y., 2023. An explainable deep learning-enabled intrusion detection framework in IoT networks. Inform. Sci. 639, 119000.

Khacha, A., Saadouni, R., Harbi, Y., Aliouat, Z., 2022. Hybrid deep learning-based intrusion detection system for industrial internet of things. In: 2022 5th International Symposium on Informatics and Its Applications. ISIA, IEEE, pp. 1–6.

Khan, M.T., Akhunzada, A., Zeadally, S., 2022a. Proactive defense for fog-to-things critical infrastructure. IEEE Commun. Mag. 60 (12), 44–49. http://dx.doi.org/10.1109/MCOM.005.2100992.

Khan, I.A., Keshk, M., Pi, D., Khan, N., Hussain, Y., Soliman, H., 2022b. Enhancing IIoT networks protection: A robust security model for attack detection in internet industrial control systems. Ad Hoc Netw. 134, 102930.
Kök, İ., Okay, F.Y., Muyanlı, Ö., Özdemir, S., 2023. Explainable artificial intelligence (xai) for internet of things: a survey. IEEE Internet Things J..
Kumar, R., Kumar, P., Tripathi, R., Gupta, G.P., Kumar, N., Hassan, M.M., 2021. A privacy-preserving-based secure framework using blockchain-enabled deep-learning in cooperative intelligent transport system. IEEE Trans. Intell. Transp. Syst. 23 (9), 16492–16503.
Li, R., Qin, Y., Wang, C., Li, M., Chu, X., 2022. A blockchain-enabled framework for enhancing scalability and security in IIoT. IEEE Trans. Ind. Inform..
Lilhore, U.K., Manoharan, P., Simaiya, S., Alroobaea, R., Alsafyani, M., Baqasah, A.M., Dalal, S., Sharma, A., Raahemifar, K., 2023. HIDM: Hybrid intrusion detection model for industry 4.0 networks using an optimized CNN-LSTM with transfer learning. Sensors 23 (18), 7856.
Lo, W., Alqahtani, H., Thakur, K., Almadhor, A., Chander, S., Kumar, G., 2022. A hybrid deep learning based intrusion detection system using spatial-temporal representation of in-vehicle network traffic. Veh. Commun. 35, 100471.
Lundberg, S.M., Lee, S.-I., 2017. A unified approach to interpreting model predictions. Adv. Neural Inf. Process. Syst. 30.
Meidan, Y., Bohadana, M., Mathov, Y., Mirsky, Y., Shabtai, A., Breitenbacher, D., Elovici, Y., 2018. N-baiot—network-based detection of iot botnet attacks using deep autoencoders. IEEE Pervasive Comput. 17 (3), 12–22.
Oseni, A., Moustafa, N., Creech, G., Sohrabi, N., Strelzoff, A., Tari, Z., Linkov, I., 2022. An explainable deep learning framework for resilient intrusion detection in IoT-enabled transportation networks. IEEE Trans. Intell. Transp. Syst. 24 (1), 1000–1014.
Rathee, G., Ahmad, F., Jaglan, N., Konstantinou, C., 2022. A secure and trusted mechanism for industrial IoT network using blockchain. IEEE Trans. Ind. Inform. 19 (2), 1894–1902.
Rawal, A., McCoy, J., Rawat, D.B., Sadler, B.M., Amant, R.S., 2021. Recent advances in trustworthy explainable artificial intelligence: Status, challenges, and perspectives. IEEE Trans. Artif. Intell. 3 (6), 852–866.
Santhadevi, D., Janet, B., 2023. HSDL-based intelligent threat detection framework for IoT network. J. Intell. Fuzzy Systems 45 (1), 1775–1790.
Shaghaghi, A., Kaafar, M.A., Buyya, R., Jha, S., 2020. Software-defined network (SDN) data plane security: issues, solutions, and future directions. In: Handbook of Computer Networks and Cyber Security: Principles and Paradigms. Springer, pp. 341–387.
Sharafaldin, I., Lashkari, A.H., Ghorbani, A.A., et al., 2018. Toward generating a new intrusion detection dataset and intrusion traffic characterization.. ICISSp 1, 108–116.
Shukla, A.K., Srivastav, S., Kumar, S., Muhuri, P.K., 2023. UInDeSI4. 0: An efficient unsupervised intrusion detection system for network traffic flow in industry 4.0 ecosystem. Eng. Appl. Artif. Intell. 120, 105848.
Tadros, C.N., Mokhtar, B., Rizk, M.R., 2023. Software defined network-based management architecture for 5g network. In: Paradigms of Smart and Intelligent Communication, 5G and beyond. Springer, pp. 171–195.
Taher, F., Abdel-salam, M., Elhoseny, M., El-hasnony, I.M., 2023. Reliable machine learning model for IIoT botnet detection. IEEE Access.
Tareq, I., Elbagoury, B.M., El-Regaily, S., El-Horbaty, E.-S.M., 2022. Analysis of ton-iot, unw-nb15, and edge-iiot datasets using dl in cybersecurity for iot. Appl. Sci. 12 (19), 9572.
Telikani, A., Shen, J., Yang, J., Wang, P., 2022. Industrial IoT intrusion detection via evolutionary cost-sensitive learning and fog computing. IEEE Internet Things J. 9 (22), 23260–23271.
Xu, P., Lu, G., Li, Y., Xu, C., 2023. EE-GCN: A graph convolutional network based intrusion detection method for iIoT. In: 2023 5th International Conference on Natural Language Processing. ICNLP, IEEE, pp. 338–344.
Zainudin, A., Ahakonye, L.A.C., Akter, R., Kim, D.-S., Lee, J.-M., 2022. An efficient hybrid-dnn for ddos detection and classification in software-defined iiot networks. IEEE Internet Things J..
Zhang, K., Yang, J., Shao, Y., Hu, L., Ou, W., Han, W., Zhang, Q., 2024. Intrusion detection model for internet of vehicles using GRIPCA and OWELM. IEEE Access.



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